Delta Lake MERGE Implementation

Week 7 Assignment Write-up

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What this assignment was about

For this week's assignment I worked across two connected pieces of work. The first half stays close to the basics — reading a CSV file into Pandas, poking around the data, cleaning it up, and writing out a clean copy. The second half moves into Delta Lake, where I looked at how incremental data gets merged into an existing table and how to keep a full history of changes using the Slowly Changing Dimension (SCD) Type 2 pattern.

The dataset I used throughout is the Sample - Superstore customer file (customer_master.csv), which I pulled in through Databricks Volumes.

Part 1 — Getting the data ready with Pandas

I started by loading customer_master.csv into a DataFrame and getting a feel for it using head(), tail(), shape, columns and dtypes — basically just looking at what I was working with before touching anything.

- Checked for missing values and filled them in with fillna() — numeric fields got 0, and the Postal Code field got 'Unknown' where it was blank.
- Filtered the data down to Region = West and pulled out a smaller set of columns just to inspect that slice.
- Ran drop_duplicates() to clear out repeated rows.
- Built a new column, total_amount, as Sales multiplied by Quantity.
- Wrote the cleaned-up version out to customer_master_cleaned.csv.

Part 2 — Merging incremental data into a Delta table

Next I moved the master dataset into a Spark DataFrame — cleaning up the column names, fixing data types, and rebuilding total_amount along the way — then saved it as a Delta table called week_7_db.customer_master.

To test a real merge scenario, I put together an incremental batch: 20 existing records with Sales bumped up by 10%, plus 10 brand-new customers, for 30 rows total. I ran this through DeltaTable.merge(), using whenMatchedUpdate() for the Row_IDs that already existed and whenNotMatchedInsertAll() for the new ones.

After the merge I double-checked the row counts before and after, made sure no duplicate Row_IDs had crept in, and looked for nulls in the key columns.

Part 3 — Adding history with SCD Type 2

The last piece was about not losing history when a record changes. I added three columns — is_current, effective_date and end_date — to the Delta table.

Instead of overwriting a row on a match, the old version now gets marked as expired (is_current = false, end_date = current_date()), and the incremental record is appended as the new 'current' row. That way both the old and new state of a changed record stick around.

The final SCD2 table was exported as customer_master_scd2.csv.

Numbers that came out of it

Pandas cleaning — before vs after

Metric	Value
Original size	9,994 rows / 21 columns
Missing values found	0 — dataset was already clean
Duplicates removed	0
Rows in the West-region filter test	3,203
Size after adding total_amount	9,994 rows / 22 columns
total_amount — avg / max / min	1,149.50 / 135,830.88 / 0.44
File written out	customer_master_cleaned.csv

MERGE (SCD Type 1) — before vs after

Metric	Value
Master rows before the merge	9,994
Incremental rows applied	30 (20 updates + 10 new)
Master rows after the merge	10,004
New rows inserted	10 — matches what was expected
Duplicate Row_IDs after merge	0
Nulls in Sales / total_amount	300 each, coming from the newly inserted test rows that had no values set

SCD Type 2 — final state

Metric	Value
Total rows across all versions	10,034
Current rows (is_current = true)	10,004
Historical / expired rows	30
Average Sales, current rows	$237.26
Total revenue, current rows	$11,542,976.19

Top region	West
Top category	Office Supplies
File written out	customer_master_scd2.csv

Where the screenshots are

All the screenshots that go with this are saved under a screenshots/ folder, split into sub-folders by stage of the work:

Folder	File	What it captures
data_loading	01_data_loaded.png	CSV loaded into a Pandas DataFrame
data_loading	02_data_explored.png	head/tail, shape, columns, dtypes output
data_loading	03_derived_column.png	total_amount column created
data_loading	04_saved_csv.png	Confirmation of the cleaned CSV export
data_cleaning	01_missing_values.png	Missing-value check output
data_cleaning	02_filter_select.png	Filtered West-region rows and selected columns
data_cleaning	03_duplicates.png	Duplicate counts before and after removal
scd1	01_master_table_created.png	Creation of the Delta master table (week_7_db.customer_master)
scd1	02_incremental_created.png	The 30-record incremental dataset being built
scd1	03_merge_execution.png	MERGE running, with row counts before/after
scd2	scd2_implementation.png	SCD2 columns added and the historical-merge logic
validation	validation.png	Row-count, duplicate, and null validation checks
final_output	01_summary.png	Final printed assignment summary
final_output	02_summary_stats.png	Avg sales, total revenue, top region/category
final_output	03_final_table_category_wise.png	Final table viewed/grouped by category

Wrapping up

Taken together, this assignment walks through a fairly complete data pipeline: cleaning raw data in Pandas, moving it into a Delta Lake table, running an ACID-safe MERGE to handle both updates and brand-new records, checking that the merge actually did what it should have, and finally layering SCD Type 2 on top so that changed records don't just get overwritten — their history sticks around too.